

Intermittent Rivers: A Challenge for Freshwater Ecology

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For many decades, river research has been focused on perennial rivers. Intermittent river research has a shorter history, and recent studies suggest that alternating dry and wet conditions alter virtually all biotic communities and biogeochemical processes in these rivers. Intermittent rivers constitute more than half of the length of the global river network and are increasing in number and length in response to climate change, land-use alteration, and water abstraction. Our views of the roles that rivers play in maintaining biodiversity and controlling material fluxes will change substantially when intermittent rivers are fully integrated into regional and global analyses. Concepts, questions, and methodologies from lotic, lentic, and terrestrial ecology need to be integrated and applied to intermittent rivers to increase our knowledge and effective management of these rivers.

Keywords: flow cessation, habitat mosaics, fragmentation, biodiversity, biogeochemistry

Rivers that naturally, periodically cease to flow are found on every continent, and these intermittent rivers may be even more common than perennial rivers. We use the term *intermittent rivers* to refer to all temporary, ephemeral, seasonal, and episodic streams and rivers in defined channels. A conservative estimate is that intermittent rivers constitute more than 30% of the total length and discharge of the global river network (Tooth 2000). For example, in Australia, roughly 70% of the 3.5 million kilometers of river channels measured at the 1:250,000 scale are considered intermittent (Sheldon et al. 2010), and more than half of the total length of rivers in the United States, Greece, and South Africa are intermittent (Larned et al. 2010). Intermittent rivers are common in Canada (Buttle et al. 2012); 25%–40% of the total length of rivers in France is intermittent (Snelder et al. 2013); and most Alpine, Arctic, and Antarctic rivers are intermittent. These estimates do not include low-order streams (i.e., headwater streams) because they are difficult to detect by traditional mapping techniques, including airborne photography and satellite imagery (Benstead and Leigh 2012). Low-order streams can make up more than 70% of river networks and are particularly prone to flow intermittence (Lowe and Likens 2005). Therefore, the proportion of intermittent rivers in the global river network is likely to be greater than 50%.

In addition to naturally intermittent rivers, water abstraction and impoundment have caused many formerly perennial rivers to become intermittent in the last 50 years, including large rivers such as the Nile, Indus, Yellow, Amu and Syr Darya, Rio Grande, and Colorado. Most of the

once-perennial rivers of arid and semiarid regions are now intermittent (Gleick 2003). In the near future, the number of intermittent rivers will increase in regions in which severe climatic drying or water appropriation occurs (Larned et al. 2010, Döll and Schmied 2012).

Intermittent rivers are a recent addition to the field of freshwater ecology. Although pioneering papers were published several decades ago (e.g., Larimore et al. 1959, Clifford 1966, Williams and Hynes 1976), they were not followed by a concerted effort to develop and advance the field of intermittent river ecology. In studies of in-stream biological communities; biogeochemical fluxes; and material exchange between rivers and the atmosphere, land, ocean, and groundwater, perennial flow and permanent surface–water connections have been presumed between river reaches. For example, according to the river continuum concept—arguably the most influential concept in river science—rivers are viewed as hydrological continua with progressive, longitudinal changes in biota and organic matter dynamics (Vannote et al. 1980). The river continuum concept and its successors (e.g., the serial discontinuity concept and the flood pulse concept; Ward and Stanford 1983, Junk et al. 1989) provide little insight into intermittent rivers, in which the complete loss of hydrological continuity affects virtually all ecological processes. In the last 20 years, intermittent river ecology has reemerged as a multidisciplinary field that links community ecology, biogeochemistry, hydrology, and river management (Larned et al. 2010). In this essay, we turn our focus to the future and the benefits and challenges of incorporating intermittent rivers into modern concepts, knowledge,

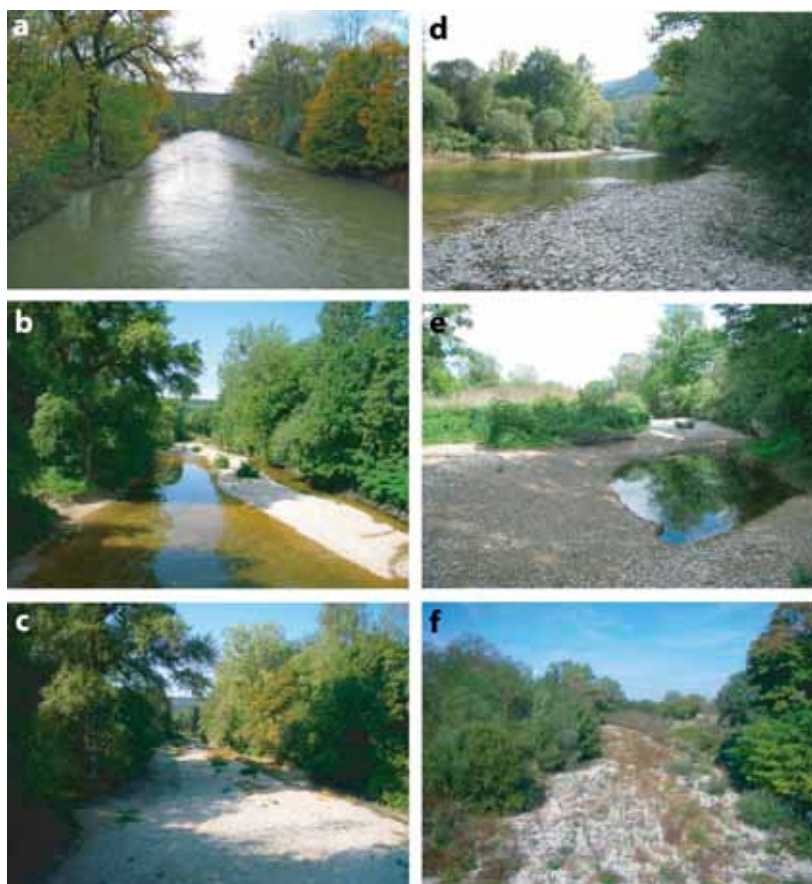


Figure 1. Shifting lotic (a, d), lentic (b, e), and terrestrial (c, f) habitats across intermittent rivers. Panels (a), (b), and (c) show a single reach along the Albarine River, in France, on different dates; panels (d), (e), and (f) show different reaches (along the same river) on a single date. Photographs: Thibault Datry (b–d) and Roland Corti (a, e, f).

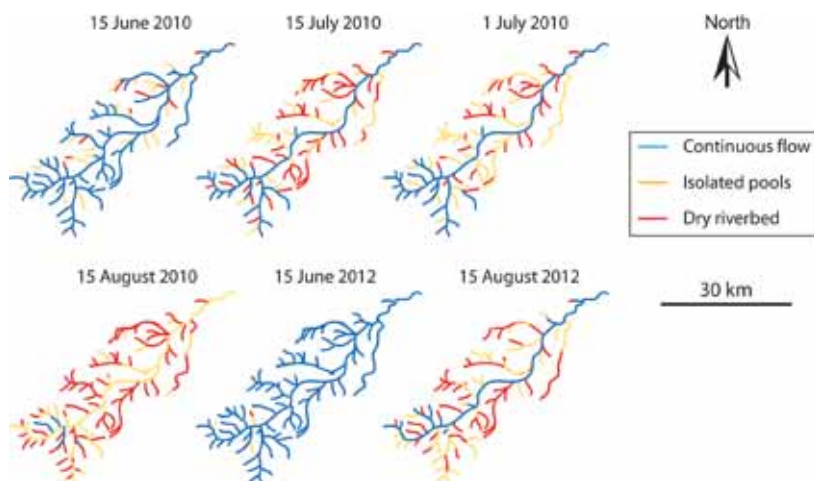


Figure 2. Expansion–contraction cycle of channel networks from the Thouaret River, in western France, showing the distribution of reaches with continuous flow (lotic, blue lines), isolated pools (lentic, orange lines), and without surface water (dry, red lines). These maps were based on fine-scale visual observations by fisherman federations of the Poitou–Charentes Region. Source: Adapted from www.eau-poitou-charentes.org. Abbreviation: km, kilometers.

and methods in freshwater and terrestrial ecology.

Shifting aquatic–terrestrial habitat mosaics

Surface-flow and groundwater-level fluctuations drive the expansion–contraction cycles of intermittent river networks (Stanley et al. 1997), which, in turn, generate alternating flowing, drying, and dry reaches. The resulting mosaic of lotic (flowing), lentic (standing), and terrestrial habitats is continuously shifting at the network scale (figures 1 and 2). The diversity, spatial arrangement, turnover, and connectivity of these habitats are controlled by the magnitude, frequency, and duration of drying events (Stanley et al. 1997, Bunn et al. 2006). Like flooding (Stanford et al. 2005, Leigh et al. 2010), drying maintains habitat heterogeneity (e.g., figure 2) and controls large-scale biodiversity and biogeochemical processes in intermittent rivers.

Intermittent rivers have long been viewed as species poor, and dry channels have been viewed as biologically inactive systems (e.g., Poff and Ward 1989, Stanley et al. 1997). However, there is growing evidence that this view is inaccurate when lentic and terrestrial habitats are taken into account. Population- and community-ecology studies of intermittent rivers are still rare, and most have been focused on lotic habitats. Generally, the lotic invertebrate communities in these studies are subsets of those in adjacent rivers or in reaches of the same river with perennial flow and are characterized by lower abundance and species richness and by truncated food webs (e.g., Williams 2006, Datry et al. 2013). Although perennial and intermittent flow regimes are often seen as categorically different, flow intermittence varies continuously, and biotic communities vary along gradients of intermittence. For example, taxonomic richness in different aquatic groups decreases continuously as the severity of intermittence increases (figure 3), which suggests a lack of clear thresholds in the response of communities to flow intermittence. Consequently, defining intermittent rivers on the basis of biota alone may be difficult or inappropriate.

Although lotic biodiversity generally decreases in response to increasing flow intermittence, increased lentic and terrestrial biodiversity can compensate for decreased

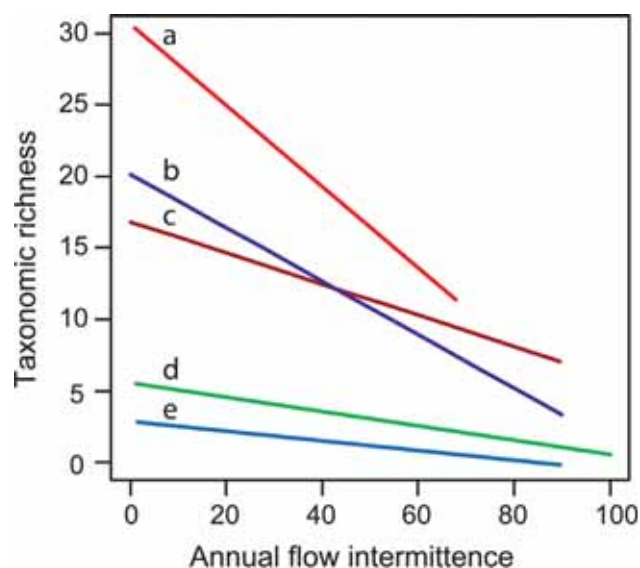


Figure 3. Relationships between annual flow intermittence (as a percentage) and the taxonomic richness (in the number of taxa) of lotic communities reported in recent studies. Annual flow intermittence is defined as the proportion of the year without surface water. The letter labels mark (a) benthic invertebrates in the Albarine River, France (Datry 2012); (b) benthic invertebrates in the Selwyn River, New Zealand (Arscott et al. 2010); (c) hyporheic invertebrates in the Selwyn River (Datry et al. 2007); (d) riparian plants in the San Pedro River, Arizona (Stromberg et al. 2005); and (e) fish in the Selwyn River (Davey and Kelly 2007). The lines are based on regression models published in the original studies.

lotic biodiversity (Larned et al. 2010). Descriptions and quantification of intermittent river biodiversity are incomplete if lotic habitats are considered exclusively. During lentic and terrestrial phases, intermittent river channels are colonized by lentic and terrestrial plants, vertebrates, and invertebrates. In disconnected, drying pools, lotic communities are gradually replaced by pond-like (lentic) and semiaquatic communities (Boulton et al. 1992, Stanley et al. 1997, Bonada et al. 2007). Diverse terrestrial arthropod communities have been reported from dry riverbeds in Europe, Australia, New Zealand, and Africa (Wishart 2000, Corti and Datry 2012, Steward et al. 2012). In arid regions, dry riverbeds are important habitats and refuges for terrestrial organisms, including large mammals (Steward et al. 2012), and the disconnected pools that persist during dry periods act as refuges for semiaquatic and aquatic organisms (Sheldon et al. 2010). In intermittent rivers, as in floodplains (Stanford et al. 2005), habitat patches shift among lotic, lentic, and terrestrial phases, and the differences in communities in each phase increase biodiversity at a larger scale. For example, 20% more riparian plant species occurred over a 2–8-year period along intermittent reaches than along

perennial reaches in three Arizona rivers (Katz et al. 2012). Similarly, the riparian zone of intermittent reaches in two French intermittent rivers contained 10%–20% more terrestrial arthropod species than did the adjacent perennial reaches over one hydrological year (the authors' unpublished data). In both cases, higher biodiversity along intermittent reaches was associated with species turnover in response to hydrological changes. In general, species in intermittent rivers are likely to account for a high proportion of regional biodiversity when the turnover rates (beta diversity) of lotic, lentic, and terrestrial communities are simultaneously considered.

Intermittent river biodiversity in a changing world

Freshwater biodiversity has been and will continue to be severely affected by global change (Dudgeon et al. 2006). However, the degree to which the biodiversity of naturally intermittent rivers is affected may differ from that of perennial rivers. Natural drying–rewetting cycles serve as evolutionary cues that increase biological, physiological, and ecological trait diversity (Williams 2006, Warfe et al. 2011), which may, in turn, enhance the adaptability of communities to future environmental change. Intermittent river species display a variety of physiological, behavioral, morphological, and life-history adaptations for surviving or exploiting dry–wet cycles. For example, estivating fish and encysting and diapausing invertebrates can survive multiple dry–wet cycles for months or years (Williams 2006, Larned et al. 2010). Conversely, some terrestrial invertebrates that inhabit intermittent rivers survive periodic inundation or use rainfall cues to avoid flash floods (Lytle and White 2007, Corti and Datry 2012). However, this pronounced adaptability of intermittent river communities to future environmental change can be counterbalanced by the strong dispersal capacity of many lotic species. High dispersal promotes gene flow between populations, which, in turn, favors evolutionarily static species (Nosil et al. 2009). Lotic invertebrates in intermittent river reaches tend to be *r* selected, with small body sizes, high fecundity, and short life cycles, which confer high passive and active dispersal potential (Williams 2006, Bonada et al. 2007). Lotic communities in intermittent river reaches are generally subsets of those from the least intermittent and perennial section; such nested communities are strongly dominated by ubiquitous species with high dispersal potential (e.g., Arscott et al. 2010, Datry et al. 2013).

Anthropogenic shifts from perennial to intermittent flow regimes may lead to irreversible modifications of biotic communities and of ecosystem processes and services. Species that are poorly adapted to dry conditions (e.g., species with low tolerance to desiccation) may be eliminated from newly intermittent rivers, and habitat fragmentation due to drying may limit recolonization (Phillipsen and Lytle 2013). In contrast, large lentic species with strong dispersal abilities and high fecundity, such as some Odonata, Hemiptera, or Coleoptera species may be minimally affected by aquatic habitat reduction and fragmentation (Bohonak and Jenkins

2003). In newly intermittent rivers, the loss of top predators, such as fish, and subsequent modifications of species interactions are expected to have cascading effects on community structure, composition, and resilience (Soulé et al. 2005). Riparian plant and animal species with high moisture requirements may be replaced by xerophilic species, and terrestrial riparian predators that rely on emerging aquatic insects may be lost. Thorough analyses of the ecological consequences of anthropogenic river drying on biodiversity will require a broader aquatic–terrestrial perspective than is currently applied in river ecology. This challenge is made greater by the fact that anthropogenic river drying often occurs in areas that are simultaneously affected by water pollution, habitat degradation, and overfishing. Many artificially intermittent rivers are characterized by hypoxia, eutrophication, and high concentrations of industrial and agricultural contaminants (e.g., Qi et al. 2005, Szykiewicz et al. 2011). These observations suggest that artificially intermittent rivers affected by multiple anthropogenic stressors will rarely match naturally intermittent rivers in terms of ecological function or biodiversity.

In contrast to the spread of artificial intermittence described above, some naturally intermittent rivers have become perennial because of controlled releases from dams and weirs; agricultural, industrial, and municipal effluent discharge; and interbasin transfers (Larned et al. 2010, Steward et al. 2012). The state of knowledge about the ecological consequences of artificial perennial flow is poor. Although flow augmentation can increase habitat availability for some lotic and lentic species, it can reduce habitat availability for terrestrial species, including some with high conservation value. Artificial perennial flow may also increase river sedimentation, channel narrowing, and vegetation encroachment (Cluett 2005) and may favor the proliferation of nonnative species, extirpate populations of native species, and reduce the availability of predator refugia (Labbe and Fausch 2000, Reich et al. 2010). In cases in which perennial flow is sustained by effluent discharge, the effects of pollutants may counteract any positive effects of reduced intermittence.

Expanding the role of rivers in biogeochemical cycles

Our understanding of the role of rivers in global and regional carbon and nutrient cycles is increasing rapidly. Rivers are longitudinal reactors that store, transform, and transfer organic matter and nutrients to adjacent floodplains, groundwater, lakes, and coasts and to the atmosphere (Battin et al. 2009, Benstead and Leigh 2012). Current large-scale assessments of biogeochemical cycling in freshwater systems rely on conservative estimates of river area and discharge, which typically omit intermittent rivers and small, low-order streams. Intermittent rivers are rarely gauged, because of their limited economic importance, and small, intermittent streams are excluded from large-scale hydrological surveys (Benstead and Leigh 2012). Estimates of material fluxes in

rivers that are based on flow and network density data for perennial rivers are underestimated, which increases the uncertainty of regional and global estimates of organic matter transformation (Battin et al. 2009), carbon dioxide efflux (Butman and Raymond 2011), and other biogeochemical processes.

To better understand the role of intermittent rivers in catchment-, regional-, and global-scale biogeochemistry, we first need to identify the differences between the organic matter dynamics and biogeochemical cycles of intermittent rivers and those of perennial rivers. In particular, the variability of organic matter and nutrient transformations may be higher and their dynamics more distinctly pulsed in intermittent than in perennial rivers. Drying and dry river channels accumulate riparian leaf litter and sediments for extended periods (Gomi et al. 2002, Acuña and Tockner 2010, Corti et al. 2011). Hypoxic conditions in disconnected pools can cause rapid ammonium and phosphate release from sediments (Von Schiller et al. 2011). As river channels dry, dissolved nutrients are converted to evaporites, and terrestrial plants gradually replace aquatic macrophytes and algae (McLaughlin 2008). During shifts to dry phases, the primary agents of organic matter degradation and mineralization shift from leaching and aquatic microorganisms and invertebrates to photodegradation and terrestrial microorganisms and invertebrates (Austin and Vivanco 2006, Corti et al. 2011). Organic matter transformation rates are very low during terrestrial phases (e.g., Stanley et al. 1997, Corti et al. 2011).

When flow resumes, the organic matter, nutrients, and terrestrial plants that have accumulated on dry channel surfaces are transported downstream by advancing wetting fronts. Depending on runoff intensity and geomorphic context, these fronts can take the form of flood bores, small rivulets, or expanding pools of upwelling groundwater (Jacobson et al. 2000, Corti and Datry 2012). Solutes and entrained organic matter at the leading edge of these wetting fronts can exceed base flow concentrations by orders of magnitude, and organic matter leaching and evaporite dissolution cause spikes in solute concentrations (Jacobson et al. 2000, Hladysz et al. 2011, Corti and Datry 2012). Advancing fronts in intermittent rivers can move many tons of organic matter to downstream receiving waters or to locations where flood attenuation results in deposition within the river channel (a video of an advancing front is available at www.irstea.fr/datry). These deposits can be important carbon sources for heterotrophic organisms (Jacobson et al. 2000, Bunn et al. 2006), but they can also cause hypoxia and fish and invertebrate kills (Hladysz et al. 2011). The ecological and societal consequences of large organic matter pulses from intermittent rivers are still unexplored, but they may affect coastal, lake, and reservoir eutrophication and hypoxia, as well as affecting greenhouse gas emissions and carbon sequestration.

Climate change and water abstraction for irrigated agriculture and other human uses are increasing the occurrence, duration, and frequency of river drying (Larned et al.

Table 1. Concepts, knowledge, and methods that are widely used in lotic, lentic, and terrestrial ecology and their potential application to intermittent river ecology and management.

Category	Example	System of origin	Application to intermittent rivers	Examples of scientific outputs	Examples of management outputs
Conceptual models	Spatial community ecology and ecosystem models	Lentic and terrestrial	Relating the changes of the habitat mosaics to community and ecosystem processes	Understanding of how lotic, lentic, and terrestrial communities are organized and interact	Guidelines for flow management in intermittent rivers
Empirical knowledge	Sampling methods for terrestrial organisms	Terrestrial	Developing sampling protocols for dry riverbeds	Description of terrestrial communities of dry riverbeds and estimates of associated biodiversity	Bioindicators of the ecological health of terrestrial phases
Technology	Lidar, remote sensing, and terrestrial laser topography	Terrestrial	Refining river drainage density by including intermittent streams	Improved estimates of biogeochemical fluxes across river networks	Large-scale maps of intermittent rivers
Predictive and simulation models	Coupled terrestrial-aquatic biogeochemical models	Lotic and terrestrial	Developing biogeochemical models that include lotic, lentic, and terrestrial phases	Biogeochemical models of intermittent rivers or reaches	Quantification of the ecosystem services provided by intermittent rivers with regard to nutrient and organic matter processing
Information synthesis	Centers for synthesis and analysis of biodiversity	Lentic, lotic, and terrestrial	Bringing together aquatic and terrestrial ecologists and biogeochemists to compile and analyze available data	Synthesis and analysis of biodiversity and biogeochemistry patterns and processes in intermittent rivers	Guidelines for flow management and biodiversity conservation in intermittent rivers

2010, Döll and Schmied 2012). In Mediterranean-climate regions of the Northern Hemisphere, climate change will also advance the season of deciduous-tree litter fall by several months (Acuña and Tockner 2010). As a result, more organic matter will accumulate in dry riverbeds, and the pulses of material transported downstream by wetted fronts may increase. These changes may increase the frequency and duration of hypoxia and fish and invertebrate kills.

Advancing the science and management of intermittent rivers

Human population growth, accelerating economic activity, land-use alteration, and climate change are all increasing the pressure on global water resources and threatening freshwater ecosystems (Dudgeon et al. 2006). Intermittent rivers are particularly endangered ecosystems, because they lack adequate management practices and protective policies and legislation. For example, recent US Supreme Court decisions questioned the protection of intermittent rivers (Leibowitz et al. 2008). In Europe, the Water Framework Directive classifications mostly ignore intermittent rivers (Logan and Furze 2002). The recent increase in intermittent river research should lead to substantial advances in knowledge and improvements in management, particularly if scientists from traditionally separated disciplines interact. Viewing intermittent rivers as shifting lotic, lentic, and terrestrial habitat mosaics puts the onus on ecologists to integrate and combine concepts and methodologies from lotic, lentic, and terrestrial ecology to understand and manage these ecosystems effectively. Table 1 provides some examples of how concepts and methodologies developed in various systems could

be applied to intermittent rivers and what basic and applied outputs could emerge from such applications.

Ecological research is undergoing a fundamental change in approaches, from the generation of primary data from small-scale experiments and surveys to the synthesis and analysis of multiple data sets from multiple sites, carried out by multidisciplinary teams using modeling and data-mining tools (Hampton et al. 2013). The newer approach is epitomized by the research carried out by working groups at centers for ecological data synthesis and analysis (e.g., the National Center of Ecological Analysis and Synthesis and the National Socio-Environmental Synthesis Center in the United States, the Synthesis Centre for Biodiversity Sciences in Germany, the Australian Centre for Ecological Analysis and Synthesis, the Centre for Ecological and Evolutionary Synthesis in Norway, the Center for Synthesis and Analysis of Biodiversity in France). At these centers, working groups use large data compilations to address ecological issues that are not amenable to traditional experiments and surveys, including continental- and global-scale patterns in ecological processes that cannot be detected by remote sensing. Many of the emerging questions in intermittent river ecology discussed in this article are well suited for the approach used at data synthesis and analysis centers. To investigate the dynamics of intermittent river populations, communities, flow regimes, and biogeochemistry and to identify climatic gradients and anthropogenic effects on these dynamics, data sets must be compiled from multiple sources, integrated, processed (e.g., standardized and georeferenced), and analyzed. Such a project intended to advance the science and management of intermittent rivers and supported

by the Center for the Synthesis and Analysis of Biodiversity in Aix-en-Provence, France, has just started through the Intermittent Rivers Biodiversity Analysis and Synthesis project (IRBAS; www.irbas.fr). The aims of IRBAS are to explore and understand biodiversity patterns in intermittent rivers, to provide policymakers and resource managers with the tools they need for effective management and restoration, and to raise awareness about the importance of these ecosystems among scientists, managers, and the public. IRBAS will notably compile and standardize the available data related to the hydrology and ecology of intermittent rivers to address, through various workshops, several questions, such as where intermittent rivers are located today and why, where they will be in the future, which components of the flow regime primarily structure aquatic and terrestrial plant and animal communities, and how terrestrial and aquatic components interact within the habitat mosaics. The data synthesis and analysis approach has proven to be tremendously productive, in terms of publications, discovery, and influence on future science (Hampton and Parker 2011), and the growing field of intermittent river ecology will likely greatly benefit from such an approach.

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References cited

- Acuña V, Tockner K. 2010. The effects of alterations in temperature and flow regime on organic carbon dynamics in Mediterranean river networks. *Global Change Biology* 16: 2638–2650.
- Arscott DB, Larned ST, Scarsbrook MR, Lambert P. 2010. Aquatic invertebrate community structure along an intermittence gradient: Selwyn River, New Zealand. *Journal of the North American Benthological Society* 29: 530–545.
- Austin AT, Vivanco L. 2006. Plant litter decomposition in a semi-arid ecosystem controlled by photodegradation. *Nature* 442: 555–558.
- Battin TJ, Luyssaert S, Kaplan LA, Aufdenkampe AK, Richter A, Tranvik LJ. 2009. The boundless carbon cycle. *Nature Geoscience* 2: 598–600.
- Benstead JP, Leigh DS. 2012. An expanded role of river networks. *Nature Geoscience* 5: 678–679.
- Bohonak AJ, Jenkins DG. 2003. Ecological and evolutionary significance of dispersal by freshwater invertebrates. *Ecology Letters* 6: 783–796.
- Bonada N, Rieradevall M, Prat N. 2007. Macroinvertebrate community structure and biological traits related to flow permanence in a Mediterranean river network. *Hydrobiologia* 589: 91–106.
- Boulton AJ, Peterson CG, Grimm NB, Fisher SG. 1992. Stability of an aquatic macroinvertebrate community in a multi-year hydrologic disturbance regime. *Ecology* 73: 2192–2207.
- Bunn SE, Thoms MC, Hamilton SK, Capon SJ. 2006. Flow variability in dryland rivers: Boom, bust and the bits in between. *River Research Application* 22: 179–186.
- Butman D, Raymond PA. 2011. Significant efflux of carbon dioxide from streams and rivers in the United States. *Nature Geoscience* 4: 839–842.
- Buttle JM, Boon S, Peters DL, Spence C, van Meerveld HJ, Whitfield PH. 2012. An overview of temporary stream hydrology in Canada. *Canadian Water Resources Journal* 37: 279–310.
- Clifford HF. 1966. The ecology of invertebrates in an intermittent stream. *Investigations of Indiana Lakes and Streams* 7: 57–98.
- Cluett L. 2005. The role of flooding in morphological changes in the regulated Lower Ord River in tropical northwestern Australia. *River Research and Applications* 21: 215–227.
- Corti R, Datry T. 2012. Invertebrates and sestonic matter in an advancing wetted front travelling down a dry river bed (Albarine, France). *Freshwater Science* 31: 1187–1201.
- Corti R, Datry T, Drummond L, Larned ST. 2011. Natural variation in immersion and emersion affects breakdown and invertebrate colonization of leaf litter in a temporary river. *Aquatic Sciences* 73: 537–550.
- Datry T. 2012. Benthic and hyporheic invertebrate assemblages along a flow intermittence gradient: Effects of duration of dry events. *Freshwater Biology* 57: 563–574.
- Datry T, Larned ST, Scarsbrook MR. 2007. Responses of hyporheic invertebrate assemblages to large-scale variation in flow permanence and surface-subsurface exchange. *Freshwater Biology* 52: 1452–1462.
- Datry T, Larned ST, Fritz KF, Bogan MT, Wood P, Meyer E, Santos A. 2013. Broad-scale patterns of invertebrate richness and community composition in temporary rivers: Effect of flow intermittence. *Ecography* 36. (7 October 2013; <http://onlinelibrary.wiley.com/doi/10.1111/j.1600-0587.2013.00287.x/abstract>) doi:10.1111/j.1600-0587.2013.00287.x
- Davey AJH, Kelly DJ. 2007. Fish community responses to drying disturbances in an intermittent stream: A landscape perspective. *Freshwater Biology* 52: 1719–1733.
- Döll P, Schmied HM. 2012. How is the impact of climate change on river flow regimes related to the impact on mean annual runoff? A global-scale analysis. *Environmental Research Letters* 7: 14–37.
- Dudgeon D, et al. 2006. Freshwater biodiversity: Importance, threats, status and conservation challenges. *Biological Reviews* 81: 163–182.
- Gleick PH. 2003. Global freshwater resources: Soft-path solutions for the 21st century. *Science* 302: 1524–1528.
- Gomi T, Sidle RC, Richardson JS. 2002. Understanding processes and downstream linkages of headwater systems. *BioScience* 52: 905–916.
- Hampton SE, Parker JN. 2011. Collaboration and productivity in scientific synthesis. *BioScience* 61: 900–910.
- Hampton SE, Strasser CA, Tewksbury JJ, Gram WK, Budden AE, Batcheller AL, Duke CS, Porter JH. 2013. Big data and the future of ecology. *Frontiers in Ecology and the Environment* 11: 156–162.
- Hladysz S, Watkins SC, Whitworth KL, Baldwin DS. 2011. Flows and hypoxic blackwater events in managed ephemeral river channels. *Journal of Hydrology* 401: 117–125.
- Jacobson PJ, Jacobson KM, Angermeier PL, Cherry DS. 2000. Variation in material transport and water chemistry along a large ephemeral river in the Namib Desert. *Freshwater Biology* 44: 481–491.
- Junk WJ, Bayley PB, Sparks RE. 1989. The flood pulse concept in river-floodplain systems. *Canadian Special Publication of Fisheries and Aquatic Sciences* 106: 110–127.

- Katz GL, Denslow MW, Stromberg JC. 2012. The Goldilocks effect: Intermittent streams sustain more plant species than those with perennial or ephemeral flow. *Freshwater Biology* 57: 467–480.
- Labbe TR, Fausch KD. 2000. Dynamics of intermittent stream habitat regulate persistence of a threatened fish at multiple scales. *Ecological Applications* 10: 1774–1791.
- Larimore RW, Childers WF, Heckrotte C. 1959. Destruction and re-establishment of stream fish and invertebrates affected by drought. *Transactions of the American Fisheries Society* 88: 261–285.
- Larned ST, Datry T, Arscott DB, Tockner K. 2010. Emerging concepts in temporary-river ecology. *Freshwater Biology* 5: 717–738.
- Leibowitz SG, Wigington PJ Jr, Rains MC, Downing DM. 2008. Non-navigable streams and adjacent wetlands: Addressing science needs following the Supreme Court's Rapanos decision. *Frontiers in Ecology and the Environment* 6: 364–371.
- Leigh C, Sheldon F, Kingsford RT, Arthington AH. 2010. Sequential floods drive “booms” and wetland persistence in dryland rivers: A synthesis. *Marine and Freshwater Research* 61: 896–908.
- Logan P, Furse M. 2002. Preparing for the European Water Framework Directive: Making the links between habitat and aquatic biota. *Aquatic Conservation: Marine and Freshwater Ecosystems* 12: 425–437.
- Lowe WH, Likens GE. 2005. Moving headwater streams to the head of the class. *BioScience* 55: 196–197.
- Lytle DA, White NJ. 2007. Rainfall cues and flash-flood escape in desert stream insects. *Journal of Insect Behavior* 20: 413–423.
- McLaughlin C. 2008. Evaporation as a nutrient retention mechanism at Sycamore Creek, Arizona. *Hydrobiologia* 603: 241–252.
- Nosil P, Harmon LJ, Seehausen O. 2009. Ecological explanations for (incomplete) speciation. *Trends in Ecology and Evolution* 23: 145–156.
- Phillipsen IC, Lytle DA. 2013. Aquatic insects in a sea of desert: Population genetic structure is shaped by limited dispersal in a naturally fragmented landscape. *Ecography* 36: 731–743.
- Poff NL, Ward JV. 1989. Implications of streamflow variability and predictability for lotic community structure: A regional analysis of streamflow patterns. *Canadian Journal of Fisheries and Aquatic Sciences* 46: 1805–1818.
- Qi F, Wei L, Jianhua S, Yonghong S, Yewu Z, Zongqiang C, Haiyang X. 2005. Environmental effects of water resource development and use in the Tarim River basin of northwestern China. *Environmental Geology* 48: 202–210.
- Reich P, McMaster D, Bond N, Metzeling L, Lake PS. 2010. Examining the ecological consequences of restoring flow intermittency to artificially perennial lowland streams: Patterns and predictions from the Broken-Boosey creek system in northern Victoria, Australia. *River Research and Applications* 26: 529–545.
- Sheldon F, Bunn SE, Hughes JM, Arthington AH, Balcombe SR, Fellows CS. 2010. Ecological roles and threats to aquatic refugia in arid landscapes: Dryland river waterholes. *Marine and Freshwater Research* 61: 885–895.
- Snelder TH, Datry T, Lamouroux N, Larned ST, Sauquet E, Pella H, Catalogne C. 2013. Regionalization of patterns of flow intermittence from gauging station records. *Hydrological Earth System Sciences* 17: 2685–2699.
- Soulé ME, Estes JA, Miller B, Honnold DL. 2005. Strongly interacting species: Conservation policy, management, and ethics. *BioScience* 55: 168–176.
- Stanford JA, Lorang MS, Hauer FR. 2005. The shifting habitat mosaic of river ecosystems. *Proceedings of the International Society of Limnology* 29: 123–136.
- Stanley EH, Fisher SG, Grimm NB. 1997. Ecosystem expansion and contraction in streams. *BioScience* 47: 427–435.
- Steward AL, von Schiller D, Tockner K, Marshall JC, Bunn SE. 2012. When the river runs dry: Human and ecological values of dry riverbeds. *Frontier in Ecology and Environment* 10: 202–209.
- Stromberg JC, Bagstad KJ, Leenhouts JM, Lite SJ, Makings E. 2005. Effects of stream flow intermittency on riparian vegetation of a semiarid region river (San Pedro River, Arizona). *River Research Application* 21: 925–938.
- Szynkiewicz A, Witcher JC, Modelska M, David MB, Pratt LM. 2011. Anthropogenic sulfate loads in the Rio Grande, New Mexico (USA). *Chemical Geology* 283: 194–209.
- Tooth S. 2000. Process, form and change in dryland rivers: A review of recent research. *Earth-Science Reviews* 5: 67–107.
- Vannote RL, Minshall GW, Cummins KW, Sedell JR, Cushing CE. 1980. The river continuum concept. *Canadian Journal of Fisheries and Aquatic Sciences* 37: 130–137.
- Von Schiller D, Acuña V, Graeber D, Martí E, Ribot M, Sabater S, Timonier X, Tockner K. 2011. Contraction, fragmentation and expansion dynamics determine nutrient availability in a Mediterranean forest stream. *Aquatic Sciences* 73: 485–498.
- Ward JV, Stanford JA. 1983. The serial discontinuity concept of lotic ecosystems. Pages 29–42 in Fontaine TD III, Bartell SM, eds. *Dynamics of Lotic Ecosystems*. Ann Arbor Sciences.
- Warfe DM, et al. 2011. The “wet-dry” in the wet-dry tropics drives river ecosystem structure and processes in northern Australia. *Freshwater Biology* 56: 2169–2195.
- Williams DD. 2006. *The Biology of Temporary Waters*. Oxford University Press.
- Williams DD, Hynes HBN. 1976. The ecology of temporary streams I. The faunas of two Canadian streams. *Internationale Revue der Gesamten Hydrobiologie und Hydrographie* 61: 761–787.
- Wishart MJ. 2000. The terrestrial invertebrate fauna of a temporary stream in southern Africa. *African Zoology* 35: 193–200.

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